



Consistency label-activated region generating network for weakly supervised medical image segmentation

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ABSTRACT

The current methods of auto-segmenting medical images are limited due to insufficient and ambiguous pathomorphological labeling. In clinical practice, rough classification labels (such as disease or normal) are more commonly used than precise segmentation masks. However, there is still much to be explored regarding utilizing these weak clinical labels to accurately determine the lesion mask and guide medical image segmentation. In this paper, we proposed a weakly supervised medical image segmentation model to directly generate the lesion mask through a class activation map (CAM) guided cycle-consistency label-activated region transferring network. Cycle-consistency enforces that the mappings between the two domains should be reversible, which ensures that the original image can be reconstructed from the translated image. We developed a complementary branches fusion module to address the issue of blurry boundaries in CAM-guided segmentation. The complementary branch preserves the original semantic information of the non-lesion region and perfectly fuses the transferred feature of the lesion region with a complementary mask-constrained fake image generation process to clear the boundary of the lesion and non-lesion regions. This module allows the class transformation to focus solely on the label-activated region, resulting in more explicit segmentation. This model can accurately identify different region of medical images at the pixel-level while preserving the overall semantic structure semantion. It organizes disease labels and corresponding regions during image synthesis. Our method utilizes a joint discrimination strategy that significantly enhances the precision of the produced lesion mask. Extensive experiments of the proposed method on BraTs, ISIC and COVID-19 datasets demonstrate superior performance over existing state-of-the-art methods. The code and datasets are available at: <https://github.com/mlcb-jlu/MedImgSeg>.

1. Introduction

Automatic medical image segmentation is an essential computer-aided diagnosis (CAD) process in various clinical applications. Profiting from the remarkable progress of deep learning techniques, automatic lesion segmentation is becoming the preferred method over traditional manual annotation due to its efficiency and consistency. Convolutional neural network (CNN)-based automatic lesion segmentation has already shown superior performance to that of physicians in specific fields such as cardiac MR segmentation [1–3] and cancerous lung nodule segmentation [4–6]. However, these outstanding achievements mainly depend on a large amount of accurately annotated ground truth which is often time-consuming and needs to be improved in clinical medical practice [7]. In medical diagnosis, the category labels of medical images

(normal or disease) are usually well-documented. Therefore, there is potential value in exploring how to leverage these labels to enhance the accuracy of medical image segmentation.

In order to tackle the issue of small-sample and less annotation of medical images, several CNN-based weakly supervised segmentation models have recently been developed [8,9]. These models use two primary strategies: Class activation map (CAM) refinement and Data augmentation. The former involves refining or constraining the rough CAM region to create a more precise segmentation mask by implementing complex network modules. The most commonly used method is a two-phase process, where a pseudo segmentation mask is generated with various CAM-correcting techniques in the first phase. For example, the non-salient region object mining method [10] uses a

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graph-based global reasoning unit to strengthen the CAM and capture global relations among disjoint and distant regions. Other methods, such as AffinityNet [11] and IRNet [12], predict the semantic affinity of CAM with random walk and detect boundaries between different object classes. In addition, Sungpil et al. introduce a texture-biased property to associate the locality of CAM [13], HRCAM extends Grad-CAM by separating the gradient maps from the class activation maps from the shallow layer for higher resolution [14]. RRM directly served as ground-truth labels for the segmentation branch, where both global information and local information sub-branches are used to generate accurate pixel-level predictions [15]. Finally, a downstream segmentation network is trained to produce the final segmentation with the pseudo mask. The second strategy involves using advanced data augmentation techniques to expand the annotated training set and improve the segmentation's robustness. Some studies, ensure consistent segmentation between the primary and auxiliary decoders, such as [16–18] generating training images with noise or using auxiliary perturbation networks, thereby improving the model's segmentation representations. In recent times, synthetic images created using generative adversarial networks [19–21] have gained popularity in augmenting data as they provide additional diversity for training data while maintaining compatible feature distribution. Moreover, transformer architecture has been introduced as the backbone network to augment the mask representation for weakly supervised segmentation, although there is a high computational cost [22].

Although the methods mentioned above have shown impressive results, they have some things that could be improved that prevent further improvements in weakly supervised medical image segmentation. For instance, the CAM refinement-based methods require a specialized transform network that can introduce additional semantic bias and harm segmentation accuracy during the second phase training process. Similarly, data augmentation-based methods only reinforce existing information and cannot provide new insights. These approaches also focus on CAM-guided pixels and struggle to identify ambiguous segmentation boundaries. Given that physicians produce a large number of medical images with image-level category labels during clinical diagnosis, it is worth exploring how these labels can be fully utilized to guide lesion segmentation in medical images without annotated segmentation masks.

In this paper, we propose a cycle-consistency label-activated region transferring network to directly generate the lesion masks for medical images with the image-level label, by extracting the sole pixel-level distinguished features initialized by the class activation map between the authentic medical images and the corresponding contrary-labeled fake images (e.g., a disease image and its corresponding synthetic or generated normal images). Cycle-consistency ensures that the original image can be reconstructed from the translated image and is used to enforce the intuition that the mappings between the two domains should be reverses of each other and that both mappings should be bijections. We consider that label transformation only on the lesion mask located region and maintain the complete semantic structure of the non-lesion region using an opposite label image synthesis framework, which can avoid pixel-level semantic inconsistency between the authentic and generated images in the non-lesion region. For this purpose, we construct a Complementary Branches Fusion Network (CBFNet) to fuse the transformation branch-derived label-distinguished lesion region and reconstruction branch-derived semantic consistency region to generate the fake images with opposite labels. The lesion mask is initialized with a class activation map (CAM) module and intensely adjusted by the transformation branch of CBFNet to capture the precise lesion boundary. Specifically, the complementary branch in CBFNet can disentangle the lesion and normal semantic region of the input image and transform the lesion region into the opposite label-indicated feature domain while maintaining the normal semantic region, which will fine-tune the precise boundary of the lesion region by contrasting and fusing the anatomical texture with the CAM-guided abnormal region. Different

from the complex two phases methods, our model is an easy-to-use one-phase network.

To verify the effectiveness of the proposed model, we apply it to BraTS [23,24], ISIC [25], and COVID-19 [26] datasets. We conducted extensive segmentation tasks, including weakly supervised segmentation with whole and few-shot samples, and found that the proposed model outperformed state-of-the-art methods. Fig. 1 shows the segmentation effect of our weakly supervised segmentation framework on BraTS dataset and ISIC dataset.

To summarize, our contribution can be outlined as follows:

- We design a one-phase weakly supervised medical image segmentation framework. This easy-to-use model produces the lesion mask by simultaneously transferring the label-activated region and maintaining the semantic region in an image synthesis network, which can avoid the unpaired images-induced semantic information inconsistency.
- The CBFNet module is proposed for adjusting lesion mask boundary, preserving semantics, and generating fake images.
- The proposed model can achieve state-of-the-art performance and present strong stability and robustness across a variety of test datasets, as well as the datasets with few samples, which is the expected property for the current small sample, unannotated medical images.

2. Related work

Weakly supervised semantic segmentation means obtaining segmentation masks based solely on image-level labels, which is crucial in addressing the issue of data scarcity. Generally, there are mainly two types of methods for weakly supervised semantic segmentation.

Discriminative Model-based. The fundamental concept behind this approach is simple and intuitive, often by refining the class-related feature heatmap, for instance, by cascading the class activation map to the random walk module. Ahn and Kwak proposed the AffinityNet to generate pixel-level pseudo labels for semantic segmentation [11]. Inspired by the ROI pooling, Liu et al. designed a multiple instance learning (MIL) frameworks to split and segment the local region of the input image using a CAM module [32]. Cooperating with the graph-based global reasoning unit, Yao et al. strengthen the classification network's ability to capture global relations among disjoint and distant regions, suppressing the intermediate feature maps of discriminative regions [10]. In addition, some methods diffuse the attention into adjacent non-discriminative parts [28,29]. The common characteristic of this method is that the extra segmentation network would be necessary for the final segmentation tasks, amplifying the inconsistent semantic introduced by the pseudo mask and leading to a significant variance in segmentation performance. Hence, the boundary of the segmentation mask obtained from those two-step methods must be clarified.

Generative Model-based. Generative Model-based. Unlike the discriminative model, the generative model often produces a high-resolution segmentation mask through an image-to-image transformation network between the different categories. A typical example is the disentangled generative model (DGM), which generates the opposite labeled image by fusing the lesion mask and input image [33]. However, the unpaired image generation will introduce some semantic background inconsistency. Another type of generative model-based segmentation is data augmentation. With various spatial transformation strategies, including scale, rotate, warp, and flip, the augmented images to help the segmentation network in a self-supervised learning mechanism [16,18,30,34]. The main area for improvement of data augmentation is that generating or augmenting images are dataset-specific and hard to transfer into new datasets. Considering the small sample and non-annotated characteristics of medical images, the above methods often prefer to capture the whole foreground or coarse-gained lesion region.

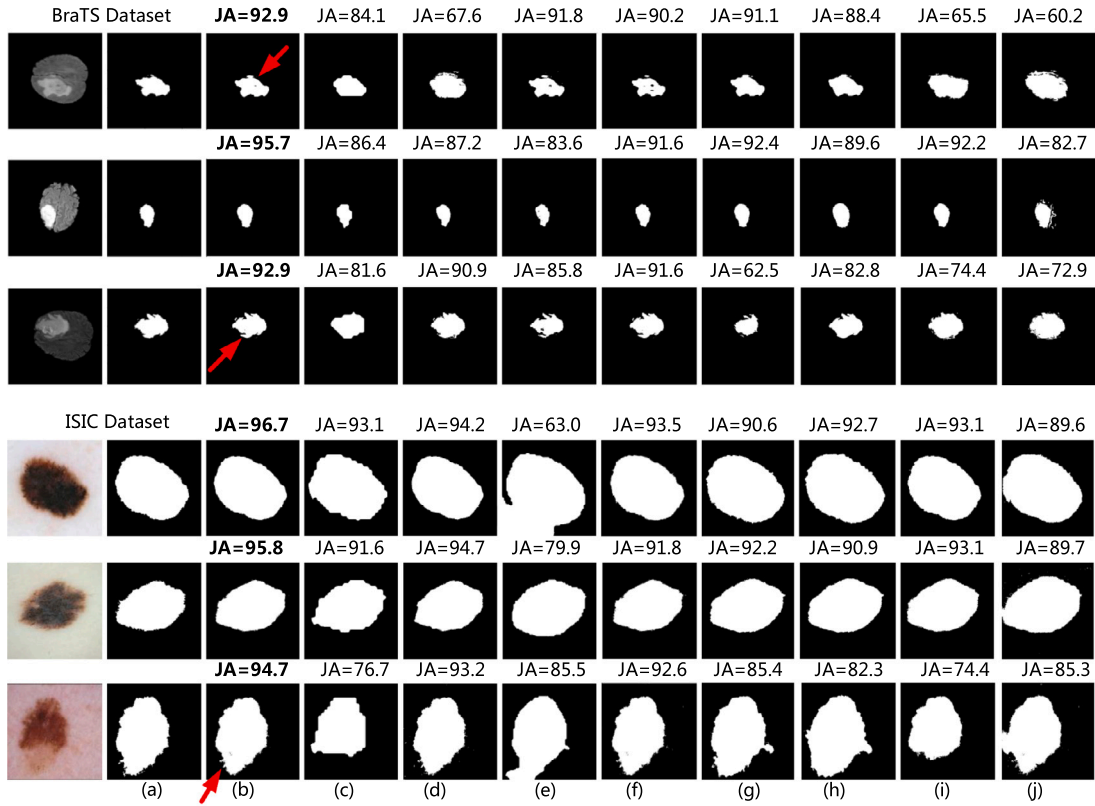


Fig. 1. Visual comparison of the semantic segmentation results between the state-of-the-art models and our proposed model. (a) Ground Truth (GT), (b) Ours, (c) PSA [11], (d) SEAM [16], (e) IRN [12], (f) AttentionGAN [27], (g) DRS [28], (h) NSROM [10], (i) RRM [29] and (j) SSeNet [30]. Red arrows indicate the ambiguous and complex lesion boundary. JA is the abbreviation for Jaccard index [31].

3. Methodology

The proposed cycle-consistency label-activated region transferring network aims to extract the lesion mask from the real disease image during the transformation process to a synthetic normal image (Fig. 2(a)). Since paired data does not exist in the real dataset, the model employs both the forward process (from disease image to normal image) and the corresponding reverse process, training under the constraint of cycle-consistency.

As illustrated in Fig. 2(a), we build an improved bidirectional label-transferring network between a disease image $x_d \in \mathbb{R}^{C \times H \times W}$ and a normal image $x_n \in \mathbb{R}^{C \times H \times W}$ to deal with the main drawback (semantic inconsistency-derived segmentation boundary blurry) of unpaired generative model-based segmentation:

(1) Unlike the traditional whole image transformation network, we utilize a CAM-based image synthesis framework to locate the label transformation only on the lesion region of a synthesized image. We achieve this by trimming and copying the non-lesion region of the input image based on the class activation mask, as shown in Fig. 2(a). This approach avoids redundant segmentation caused by inconsistencies between input and generated images, commonly occurring with traditional GAN techniques. Our method significantly reduces training costs, making it well-suited for small sample medical image segmentation.

(2) The canonical CAM-based lesion mask often degenerates into a relatively smooth region which usually exceeds the actual lesion boundary in practice. To address this issue, we design the CBFNet to solve this problem, which preserves the original semantic information of the non-lesion region and perfectly fuses the transferred feature of the lesion region with a complementary mask-constrained fake image generation process to clear the boundary of the lesion and non-lesion regions, as shown in Fig. 2(b).

3.1. Generator with label transformation network

We design our label transformation network $G_{X_d \rightarrow X_n}$ as a structure sequentially concatenated by encoder, CAM module, CBFNet and decoder (see Fig. 2(a)). In detail, the feature map f_1 , extracted from x_d by the encoder, is first sent into the CAM module to produce a label activation map f_2 . Together, f_1 and f_2 are the input of CBFNet, and there are three sub-branches in the CBFNet. For the mask branch, CBFNet generates a lesion mask feature map e_1 from input f_2 , and then e_1 is decoded as the lesion mask m , which is served as the final segmentation and guides the fake image synthesis (see Section 3.2 for the detail). To make sure e_1 captures the feature map of the lesion region, CBFNet produces a feature map e_2 with a transformation branch, which can transform the e_1 indicated lesion region into the opposite label-derived feature domain and smooth the color texture and semantic information of the fusion boundary between the e_1 indicated lesion region and non-lesion region (see the details in Section 3.2). CBFNet also contains the third branch to preserve the original semantic information of non-lesion regions for input images with a semantic feature map e_3 . Furthermore, the fused feature map e_2 is then decoded into the generated image:

$$x_{n1} = G_{X_d \rightarrow X_n}(x_d)', \quad (1)$$

where $G_{X_d \rightarrow X_n}(\cdot)'$ represents the generated image of $G_{X_d \rightarrow X_n}$. To further fine-tune the pixel-level lesion mask, we add an image synthesis layer using the transformed image x_{n1} and lesion mask m (see Fig. 2(a)):

$$x_{n2} = G_{X_d \rightarrow X_n}(x_d)'' = x_d \times (1 - m) + x_{n1} \times m, \quad (2)$$

where $G_{X_d \rightarrow X_n}(\cdot)''$ represents the synthetic images of $G_{X_d \rightarrow X_n}$. m and $1 - m$ represent the lesion mask and the normal mask, respectively. To avoid non-smooth connections between the lesion and non-lesion regions, we use both the lesion mask and the normal mask to derive the synthetic image. In addition, the reconstruction image x_{nr} , generated

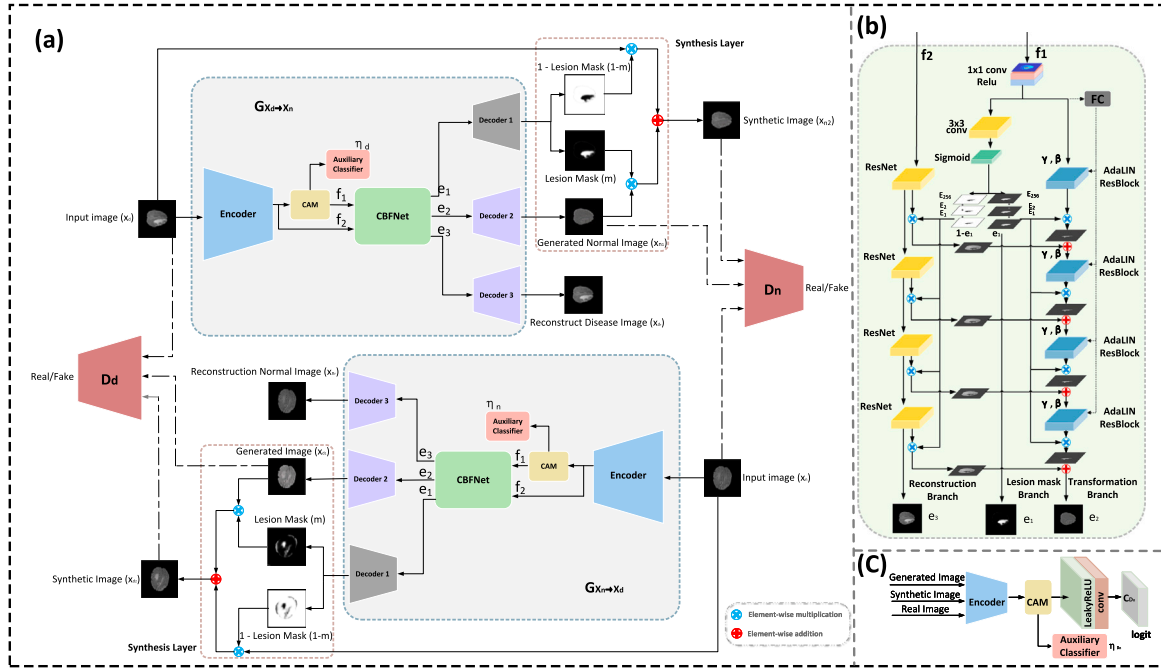


Fig. 2. (a) Overview of our cycle-consistency label transferring network. (b) The architecture of CBFNet containing three branches. (c) The architecture of the multi-scale discriminator.

from e_3 , is used to maintain the semantic feature consistency during the transformation. Both x_{n1} and x_{n2} are optimized by the discriminator D_n , while the reconstruction quality of x_{d1} is evaluated by the MSE loss between x_d and x_{d1} . Then we use the Dense CRF [35] with the parameter alpha to process the generated lesion mask obtained by our model at the test stage. Here, we only explain $G_{X_d \rightarrow X_n}$ and D_n (Section 3.3) as the vice versa should be straight-forward.

3.2. Complementary branches fusion network

We design a Complementary Branches Fusion Network (CBFNet) to disentangle the lesion and normal semantic region of the input image and transform the lesion region into the opposite label-indicated feature domain while maintaining the normal semantic region, which will fine-tune the precise boundary of the lesion region by contrasting and fusing the anatomical texture with the CAM-guided abnormal region (see Fig. 2(b)). It contains three branches, the lesion mask branch for guiding the feature fusion, the reconstruction branch for preserving semantic information, and the transformation branch for generating images with opposite categories.

Reconstruction Branch. Specifically, CBFNet receives f_1 , the complete semantic feature map of x_d , as the input for the reconstruction branch. Furthermore, the output feature map of each reconstruction layer, e_3 , mainly worked as one of the inputs of the corresponding fusion feature layer in the transformation branch. The final output of the reconstruction branch is used to maintain the original semantic information by reconstructing the input image using four cascaded convolution layers.

Lesion Mask Branch. CBFNet receives f_2 , the label-activated feature map initialized by CAM, as the input for the lesion mask branch. The lesion mask branch contains convolution and sigmoid activation layers to generate the initialized lesion mask feature map e_1 . Then, the transformation branch optimizes this feature map to identify the precise lesion boundary.

Transformation Branch. CBFNet also receives f_2 as the input for the transformation branch. Then f_2 is initially transformed into the opposite label-indicated feature map e_2 with the Adaptive Layer-Instance

Normalization module (AdaLIN [36]):

$$e_2 = \gamma \cdot (\rho \cdot \hat{f}_{1I} + (1 - \rho) \cdot \hat{f}_{1L}) + \beta, \quad (3)$$

$$\hat{f}_{1I} = \frac{f_2 - \mu_I}{\sqrt{\sigma_I^2 + \epsilon}}, \quad \hat{f}_{1L} = \frac{f_2 - \mu_L}{\sqrt{\sigma_L^2 + \epsilon}},$$

where the module parameters μ_I , μ_L and σ_I , σ_L are channel-wise, layer-wise mean and standard deviation of f_2 , respectively, γ and β are dynamically computed by a fully connected layer from the f_2 . The processing of ρ is the same as existing study [36]. In order to adjust the precise boundary of lesion mask m and generate a semantic consistency fake image with the input image (exclude the lesion mask region), we consider the following feature fusion process for each transformation layer:

$$e_2 = e_2 \times e_1 + e_3 \times (1 - e_1). \quad (4)$$

This formula means the transformed feature map e_2 mainly concentrates on the lesion mask e_1 indicated region. In contrast, the transformed feature map in $1 - e_1$ region is directly copied from e_3 to preserve the critical semantic information of the input image. This fusion process is repeated in the transformation branch four times to relax the feature inconsistency from two complementary branches.

Finally, the produced feature maps e_1 , e_2 , and e_3 generate the lesion mask, transformed image, and reconstructed image.

3.3. Discriminator

Let $x_n \in X_n$, $x_{n1}, x_{n2} \in G_{X_d \rightarrow X_n}(x_d)$ represent a group of samples from the normal image dataset, the matched generated normal image, and the synthetic normal image corresponding to the input disease image x_d . Then the discriminator D_n is a multi-scale network consisting of an encoder, a classifier C_{D_n} , and an auxiliary classifier η_{D_n} (see Fig. 2(c)). Finally, the C_{D_n} is trained to discriminate x_n as a real image, x_{n1} and x_{n2} as fake images, while η_{D_n} classifies those three images into normal images.

3.4. Loss function

The total loss function consists of five parts. Let $G_{X_d \rightarrow X_n}(\cdot)'$ and $G_{X_n \rightarrow X_d}(\cdot)'$ represent the generated image of $G_{X_d \rightarrow X_n}$ and $G_{X_n \rightarrow X_d}$. Similarly, let $G_{X_d \rightarrow X_n}(\cdot)''$ and $G_{X_n \rightarrow X_d}(\cdot)''$ represent the synthetic image of $G_{X_d \rightarrow X_n}$ and $G_{X_n \rightarrow X_d}$, given the image $x_d \in X_d$ and $x_n \in X_n$, each loss is defined as follows:

(1) After the label transformation of x_d from X_d to X_n and from X_n to X_d , the image should be successfully translated back to the original domain. Also, for x_n . The cycle-consistency loss, which is calculated as follows, ensures the translated image can be transformed back into the original image and is used to enforce the idea that the mappings between the two domains should be reverse of each other, and both mappings should be one-to-one correspondences.

$$\mathcal{L}_{cyc} (G_{X_d \rightarrow X_n}, G_{X_n \rightarrow X_d}) = \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left\| G_{X_n \rightarrow X_d} (G_{X_d \rightarrow X_n} (x_d)')' - x_d \right\|_1 \right] + \mathbb{E}_{x_n \sim p_{data}(x_n)} \left[\left\| G_{X_d \rightarrow X_n} (G_{X_n \rightarrow X_d} (x_n)')' - x_n \right\|_1 \right]. \quad (5)$$

(2) For the image transformation process from X_d to X_n , the adversarial loss can be formulated as follows:

$$\mathcal{L}_{GAN} (G_{X_d \rightarrow X_n}, D_n) = \mathbb{E}_{x_n \sim p_{data}(x_n)} [D(x_n)^2] + \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left(1 - D(G_{X_d \rightarrow X_n}(x_d)') \right)^2 \right] + \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left(1 - D(G_{X_d \rightarrow X_n}(x_d)')' \right)^2 \right]. \quad (6)$$

(3) We also add the identity loss [37] in our model to ensure that the transformation network does not change the semantic identity of the input data:

$$\mathcal{L}_{idt} (G_{X_d \rightarrow X_n}, G_{X_n \rightarrow X_d}) = \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left\| G_{X_n \rightarrow X_d}(x_d)' - x_d \right\|_1 \right] + \mathbb{E}_{x_n \sim p_{data}(x_n)} \left[\left\| G_{X_d \rightarrow X_n}(x_n)' - x_n \right\|_1 \right]. \quad (7)$$

(4) By exploiting the differential information from the auxiliary classifiers η_d , η_n , η_{D_n} , and η_{D_d} , the generator and the discriminator get to know where they need to improve or what makes the most difference between two label domains (which will result in a label activation map):

$$\mathcal{L}_{cam}(\eta_d) = - \left(\mathbb{E}_{x_d \sim p_{data}(x_d)} [\log(\eta_d(x_d))] + \mathbb{E}_{x_n \sim p_{data}(x_n)} [\log(1 - \eta_d(x_n))] \right), \quad (8)$$

$$\mathcal{L}_{cam}(G_{X_d \rightarrow X_n}, \eta_{D_n}) = \mathbb{E}_{x_n \sim p_{data}(x_n)} \left[\left(\eta_{D_n}(x_n) \right)^2 \right] + \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left(1 - \eta_{D_n}(G_{X_d \rightarrow X_n}(x_d)') \right)^2 \right]. \quad (9)$$

(5) The reconstruction branch can maintain the semantic structure of the input image and help the lesion mask cover the lesion more accurately. Let x_{dr} and x_{nr} be the output of the reconstruction branch, the reconstruction loss is calculated as follows:

$$\mathcal{L}_{recon}(x_d, x_n) = \mathbb{E}_{x_d \sim p_{data}(x_d)} \left[\left\| x_d - x_{dr} \right\|_1 \right] + \mathbb{E}_{x_n \sim p_{data}(x_n)} \left[\left\| x_n - x_{nr} \right\|_1 \right]. \quad (10)$$

Finally, our complete loss function $\mathcal{L}$ is formed by all the loss functions:

$$\mathcal{L} = \lambda_{GAN} \mathcal{L}_{GAN} + \lambda_{cyc} \mathcal{L}_{cyc} + \lambda_{idt} \mathcal{L}_{idt} + \lambda_{cam} \mathcal{L}_{cam} + \lambda_{recon} \mathcal{L}_{recon}. \quad (11)$$

4. Experiments and results

4.1. Datasets and evaluation metrics

To evaluate the segmentation performance of different methods, we conduct experiments on three medical datasets, including the BraTS dataset, ISIC dataset, and COVID-19 dataset.

BraTS dataset: BraTS 2018 [23] and BraTS 2019 [24] data are utilized in our experiments. The image data of these datasets contain 155 2D slices for each of 527 patients. In order to adapt to the network architecture, we filter 2D slices by removing the slices in which the brain area is less than 1/4 whole image. Finally, we randomly divide the dataset to obtain 563, 90, and 90 training, validation, and testing data for each cross-validation experiment.

ISIC dataset: The ISIC dataset in our experiments is obtained from ISIC 2018 dataset [25], which contains data for the skin lesion segmentation challenge. The image size ranges from 771×750 to 6748×4499 . We randomly divide the dataset to obtain 200, 31, and 31 training, validation, and testing data for each cross-validation experiment.

COVID-19 dataset: The dataset also comprises 2D slices CTs sourced from Radiopaedia [26]. Our pre-processing refers to the BraTS dataset. Finally, we randomly divide the dataset to obtain 155, 25, and 25 training, validation, and testing data for each cross-validation experiment.

During the training phase, we only use image-level labels (class labels) for weakly supervised image segmentation. In order to reduce the influence of random selection, we repeat the data division process five times and obtain five sets of experimental data for each of the three datasets. All results of the following experiments are obtained by running five times on the five experimental data sets. Furthermore, to evaluate the segmentation performance of different methods, we use the Jaccard index (JA), pixel-wise accuracy (AC), and dice coefficient (DI) [31] to measure the segmentation performance.

4.2. Implementation details

The network architectures in Fig. 2 are implemented with pytorch. In detail, the generator encoder comprises two down-sampling layers with a stride size of 2, and three decoders consist of two up-sampling convolution layers with a stride size of 1. The CBFNet comprises four AdaLIN residual blocks [36] in the transformation branch and four residual blocks in the reconstruction branch. We use the Adam optimizer with a learning rate of 0.0002, beta1 of 0.5, and beta2 of 0.999 to train the generator and the discriminator. For data augmentation, we flipped the images horizontally with a probability of 0.5, resized them to 286×286 , and randomly cropped them to 256×256 . In addition, we use a random contrast with a contrast floating range of 0.9 to 1.1. And we set $\lambda_{GAN} = 1$, $\lambda_{cyc} = 10$, $\lambda_{idt} = 10$, $\lambda_{cam} = 1000$ and $\lambda_{recon} = 10$ in Eq. (11). All experiments are done on a workstation with an Ubuntu 18.04.2 LTS system equipped with NVIDIA TITAN RTX GPU. The model's parameters are 105M and occupy approximately 11 g of memory.

4.3. Segmentation results

Weakly Supervised Image Segmentation. We generate the lesion segmentation masks using our model on BraTS and ISIC datasets to evaluate the performance of weakly supervised image segmentation. Firstly, we compare the segmentation performance of our weakly supervised segmentation model with the classic the supervised segmentation baseline UNet++ [38] using different amounts of training data. In Fig. 3, it can be observed that as the number of training samples increases, there is a steady improvement in the segmentation performance of our proposed model and the baseline UNet++. Furthermore, for different amounts of training data, the segmentation performance of our weakly supervised segmentation model is comparable to the performance of supervised segmentation baseline, especially in the small sample cases. Another observation is that the performance variance of our method is decreased with the incremental training samples.

Then, we further compare the segmentation performance of our model with other state-of-the-art weakly supervised segmentation models on BraTS and ISIC datasets. We classify these weakly supervised segmentation models into four categories based on their characteristics. As shown in Table 1, the letter G means whether the models are based

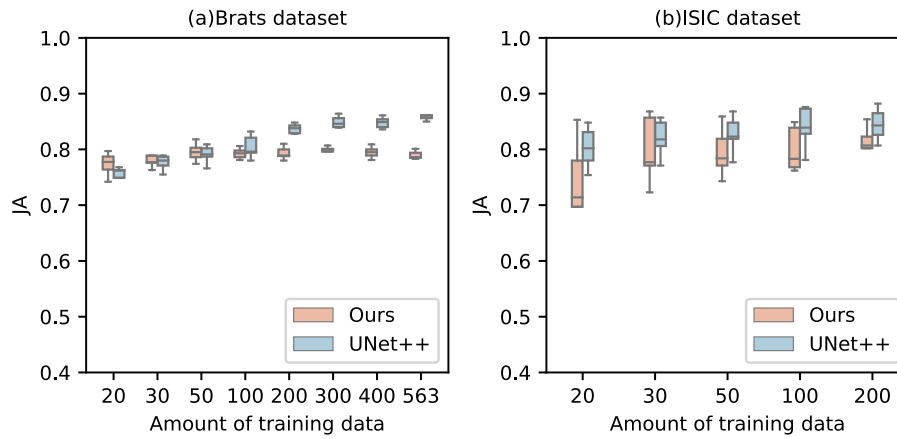


Fig. 3. The segmentation performance JA (%) of supervised baseline and our model with different amount of training data on (a) BraTS dataset and (b) ISIC dataset.

Table 1

The comparison of the segmentation performance of our model on BraTS dataset and ISIC dataset against other weakly supervised models.

Method	G	C	A	E	JA(BraTS)	AC(BraTS)	DI(BraTS)	JA(ISIC)	AC(ISIC)	DI(ISIC)
PSA [11]	–	✓	–	–	71.8±2.27	98.9±0.27	82.6±1.70	80.2±3.75	93.0±2.71	87.7±2.91
SEAM [16]	–	✓	–	–	75.2±3.01	99.0±0.30	84.0±2.97	75.5±5.27	92.0±2.14	83.9±4.07
IRN [12]	–	✓	–	–	68.3±1.16	98.9±0.04	77.9±1.30	77.5±3.72	93.6±0.95	85.9±3.35
AttentionGAN [27]	✓	–	✓	✓	72.2±3.26	98.9±0.26	80.5±3.23	79.8±3.83	94.6±1.04	87.1±2.55
DRS [28]	–	✓	–	–	73.8±5.53	99.1±0.16	82.1±5.32	74.9±5.70	92.8±1.93	84.5±3.99
NSROM [10]	–	✓	–	–	69.7±7.31	98.5±0.97	79.9±5.76	76.6±5.01	93.3±1.72	86.0±3.47
RRM [29]	–	✓	–	–	72.0±2.45	98.6±0.37	81.2±2.11	80.0±2.65	94.9±0.87	87.5±2.40
SSENet [30]	–	✓	–	–	69.1±4.70	98.1±0.55	80.1±3.72	79.1±5.86	93.7±1.88	86.8±4.53
WeakTr [39]	–	✓	–	–	76.5±1.60	99.1±0.19	85.5±1.27	80.7±5.34	95.2±1.34	87.7±4.30
Ours	✓	✓	✓	✓	79.0±0.67	99.2±0.07	87.3±0.50	81.8±1.97	95.4±0.47	88.9±1.64

Table 2

The comparison of the segmentation performance of our model with few training samples on BraTS dataset and ISIC dataset against other weakly supervised models.

Method	JA(BraTS)	AC(BraTS)	DI(BraTS)	JA(ISIC)	AC(ISIC)	DI(ISIC)
PSA [11]	64.5 ±5.52	98.3 ±0.71	76.1 ±5.04	75.9±2.60	89.1±2.71	84.1±2.35
SEAM [16]	52.3 ±5.05	96.6 ±0.84	60.9 ±6.05	74.8±2.70	92.6±1.43	83.7±3.11
IRN [12]	61.9 ±2.89	98.7 ±0.15	71.5 ±3.05	72.6±5.74	90.6±3.95	81.9±4.65
AttentionGAN [27]	68.8 ±1.63	98.6 ±0.51	77.5 ±1.35	75.5±7.37	93.3±2.39	82.9±6.95
RRM [29]	67.3 ±2.89	98.2±0.62	77.9±2.60	76.6±3.58	94.0±0.90	85.1±3.15
SSENet [30]	44.9 ±2.37	91.0±0.70	55.2±2.13	77.6±5.19	92.6±2.91	86.0±4.16
WeakTr [39]	74.6 ±1.95	98.8±0.30	83.8±1.71	79.8±4.31	94.6±1.08	87.1±3.57
Ours	79.5±1.49	99.2±0.17	87.7±1.21	78.3±2.83	94.3±0.74	86.3±2.09

on deep generative architecture. The letters C and A indicate that these models contain a CAM module and an attention mask module, respectively. The letter E means the models are learning in an end-to-end way. It is worth noting that among these models, except for our model and AttentionGAN [27], the other models use the multi-stage training strategy (against the end-to-end).

As shown in Table 1, the performance of our model on BraTS and ISIC datasets shows significant improvement compared to other models of weakly supervised segmentation. Our model achieves 79.0% and 81.8%, surpassing other best models by 2.5% and 1.1% on JA, respectively. The experimental results show that when only using the labels of image-level, our weakly supervised segmentation model can gain an available improvement.

Weakly Supervised Segmentation with Few Samples. To further mine the proposed model's segmentation ability, we compare our model's segmentation performance with state-of-the-art models of weakly supervised segmentation on datasets with few samples, which is a common situation in clinical practice. We randomly select 50 samples from the well-built training data used in the above section to train the models of weakly supervised segmentation.

Table 2 compares the segmentation performance of our model against other weakly supervised models on BraTS and ISIC datasets

with few samples. Specifically, Table 2 presents that our model achieves a 79.5% JA on BraTS dataset, surpassing other best models by 4.9%. As well as, it is observed that our model achieves 78.3% on JA on ISIC dataset, which is close with the result by WeakTr. The results show that our model can achieve state-of-the-art segmentation performance on smaller medical image data sets and is comparable to the complete training set segmentation. Contrastively, compared methods' performance is decreased on the BraTS dataset; even the DRS and NSROM fail to identify a rational segmentation mask. So, our model can be applied to medical image datasets with few samples. We also notice that although there is a higher average JA score on small samples against the complete samples on BraTS dataset, as expected, the performance variance is also more significant across small samples training data.

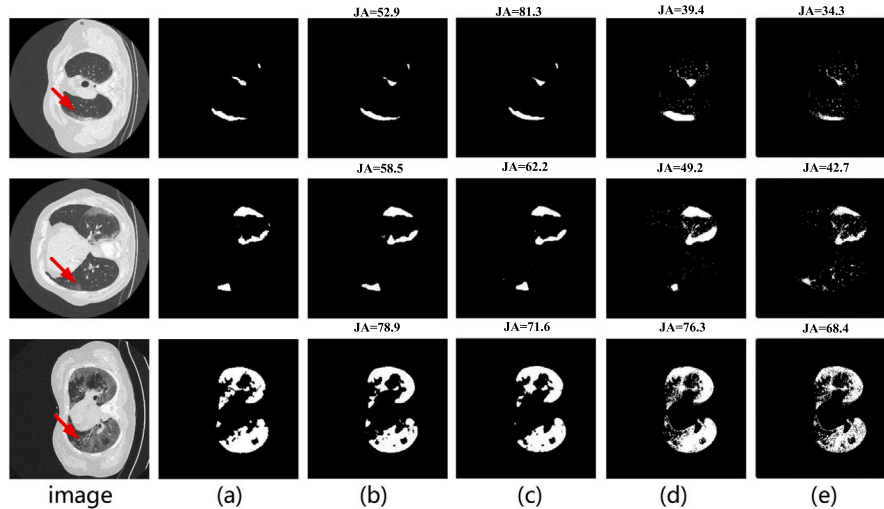
4.4. Ablation studies

In this section, we explore the influence of each designed component in our model on the segmentation performance. Furthermore, to validate our proposed CBFNet, the synthesizing of fake image (SFI) and reconstruction branch (RB) play the dominant role in improving the performance of our model. The detailed analysis of each module is as follows:

Table 3

The segmentation performance with the ablation of different component on the BraTS and ISIC datasets.

Method	JA(BraTS)	AC(BraTS)	DI(BraTS)	JA(ISIC)	AC(ISIC)	DI(ISIC)
w/o CBFNet	61.3±8.46	97.9±1.05	73.5±7.75	62.6±5.17	89.5±1.47	74.6±5.47
w/o RB	61.6±3.86	98.0±0.44	74.7±2.83	79.2±5.71	93.9±2.01	87.2±4.13
w/o SFI	51.5±9.54	92.5±5.90	64.2±9.24	65.8±7.97	90.4±2.23	77.6±6.29
w/o GFI	78.9±1.12	99.2±0.16	87.2±0.88	79.9±3.20	94.8±0.99	88.0±2.34
w/o CCL	78.1±4.51	99.1±0.57	86.7±3.99	59.9±24.70	75.0±25.18	69.0±19.19
w/o CAM	77.4±0.94	99.2±0.07	86.4±0.67	80.3±4.56	94.9±1.27	88.2±3.24
CAM	51.0±10.55	98.3±0.31	47.9±24.5	48.8±1.78	87.4±0.64	63.2±1.82
Full modules	79.0±0.67	99.2±0.07	87.3±0.50	81.8±1.97	95.4±0.47	88.9±1.64

**Fig. 4.** Comparison of our model and supervised baseline on COVID-19 dataset. (a) Ground Truth (GT), (b) Supervised baseline UNet++ [38], (c) Ours enhanced by few shot (50 samples) learning, (d) Ours, (e) AttentionGAN [27]. Red arrows indicate the ambiguous and complex lesion boundary.

Complementary Branch Fusion Network (CBFNet). One of the essential components of our model is the Complementary Branch Fusion Network (CBFNet) module, which determines the clear segmentation boundary. Here we compare the segmentation performance of the model with and without the CBFNet module. As shown in Table 3, the CBFNet module can improve the segmentation performance of 17.7% and 19.2% of JA on the BraTS and ISIC datasets, respectively.

Reconstruction Branch (RB). The reconstruction branch can maintain the semantics information so that the semantics information will be inherited during the image translation process, which promotes lesion mask boundary to have higher accuracy. Table 3 shows the segmentation performance of the reconstruction branch. By using the reconstruction branch, the segmentation performance of JA is improved 18% and 2.6% on BraTS and ISIC datasets, respectively.

The Synthetic Fake Image (SFI). To make the lesion mask more accurate, we use the generated image, lesion mask, and the input image to synthesize a fake image and feed it into the discriminator. The effect of the synthesizing image on the segmentation performance is shown in Table 3. With the synthesizing of fake images, the segmentation performance of JA has been improved by 28.1% and 16%, respectively.

The Generated Fake Image (GFI). When we do not discriminate the generated false images, the segmentation performance decreases by 0.1% and 1.9%, respectively. The result shows that the discrimination of the generated fake image can slightly improve the segmentation performance.

Cycle-consistency Loss (CCL). When we do not use cycle-consistency loss, the segmentation performance decreases by 0.9% and 21.9%, respectively. The results validate that cycle-consistency has a discernible impact on the outcomes across different datasets.

The Class Activation Map (CAM). To validate our generative model essentially improve the CAM-based coarse mask, we first evaluate the segmentation performance only using the CAM module. The

JA score of segmentation directly generated by CAM is only 51.0% and 48.8% on BraTS and ISIC datasets, respectively, which are 28% and 33% lower than our model. Then, we evaluate the segmentation performance of our model without the CAM module. After removing the CAM module, the segmentation performance of the model decreases by 1.6% and 1.5%, respectively, which shows that the CAM module cannot effectively promote the segmentation performance. Furthermore, designed components in our model play a dominant role in improving the performance of the CAM-based mask and achieving better segmentation performance.

4.5. Segmentation on CT data of COVID-19

We also explore the application of our model on CT data of COVID-19. The sample sizes of training, validation, and test set are 155, 25, and 25 in the weakly supervised experiments, respectively. For semi-supervised and supervised experiments, we used the same validation and test sets as in the weakly supervised experiments, with 50 samples selected from the complete training set as image-level labeled samples. Additionally, we used the remaining 105 samples as category-only labeled samples for semi-supervised experiments. As shown in Fig. 4, our segmentation result visualization is quite close to the result of the supervised segmentation baseline as well as ground truth. Interestingly, when applied to the few-shot learning with Unet++ [38], our weakly supervised model achieves comparable performance to the supervised baseline. The segmentation of our semi-supervised model outperforms the supervised model Unet++ by 0.7%. Contrastively, another weakly supervised method AttentionGAN [27] fails to identify a rational lesion region due to the ambiguous and complex lesion boundary (see Fig. 4 and Table 4).

Table 4

The comparison of the segmentation performance of our model on COVID-19 dataset against other methods.

Method	JA	AC	DI
Unet++(Supervised)	76.0 $\pm$ 2.81	99.2 $\pm$ 0.07	86.0 $\pm$ 1.73
Ours(Semi-supervised)	76.7$\pm$2.46	99.2 $\pm$0.07	86.4 $\pm$1.71
AttentionGAN(Weakly-supervised)	41.3 $\pm$ 2.81	97.2 $\pm$ 0.78	55.7 $\pm$ 2.62
Ours (Weakly-supervised)	61.0 $\pm$3.53	98.5 $\pm$0.17	74.6 $\pm$3.14

5. Conclusion

In order to alleviate the problem that medical image data pixel-level annotation needs medical expert knowledge and acquisition cost is high, the application of weakly supervised semantic segmentation technology to medical image segmentation task has high research significance and application value. In this paper, we propose a cycle-consistency label-activated region transferring network to refine the lesion mask from only the category label of medical images, by the proposed complementary branch fusion network. With extensive experiments on multi-datasets for weakly supervised segmentation, we confirm the effectiveness and robustness of our method. Although the transformer based model, such as MCTformer [40] and MCT-former+ [41], has inclined to be the preference of the segmentation task, we claim here that the generative model presents a comparative performance with a rather low computation cost.

Although our model has achieved good segmentation performance on experimental datasets, some limitations still need to be solved in future work. Our model currently cannot accomplish multi-class and multi-objective medical image segmentation, representing many clinical diagnoses. In future work, we will improve our model to achieve better segmentation accuracy and apply our framework to 3D medical image segmentation.

CRedit authorship contribution statement

Wei Du: Writing – review & editing, Validation, Supervision, Conceptualization. **Yongkang Huo:** Writing – review & editing, Validation, Project administration, Formal analysis. **Rixin Zhou:** Writing – original draft, Visualization, Resources, Methodology, Formal analysis. **Yu Sun:** Methodology, Formal analysis, Data curation. **Shiyi Tang:** Writing – review & editing, Formal analysis. **Xuan Zhao:** Writing – review & editing. **Ying Li:** Writing – review & editing, Supervision, Formal analysis. **Gaoyang Li:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2024.108380>.

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